Letrozole cohort: full-history NMF (k=7) vs windowed ±18 mo NMF (k=13)

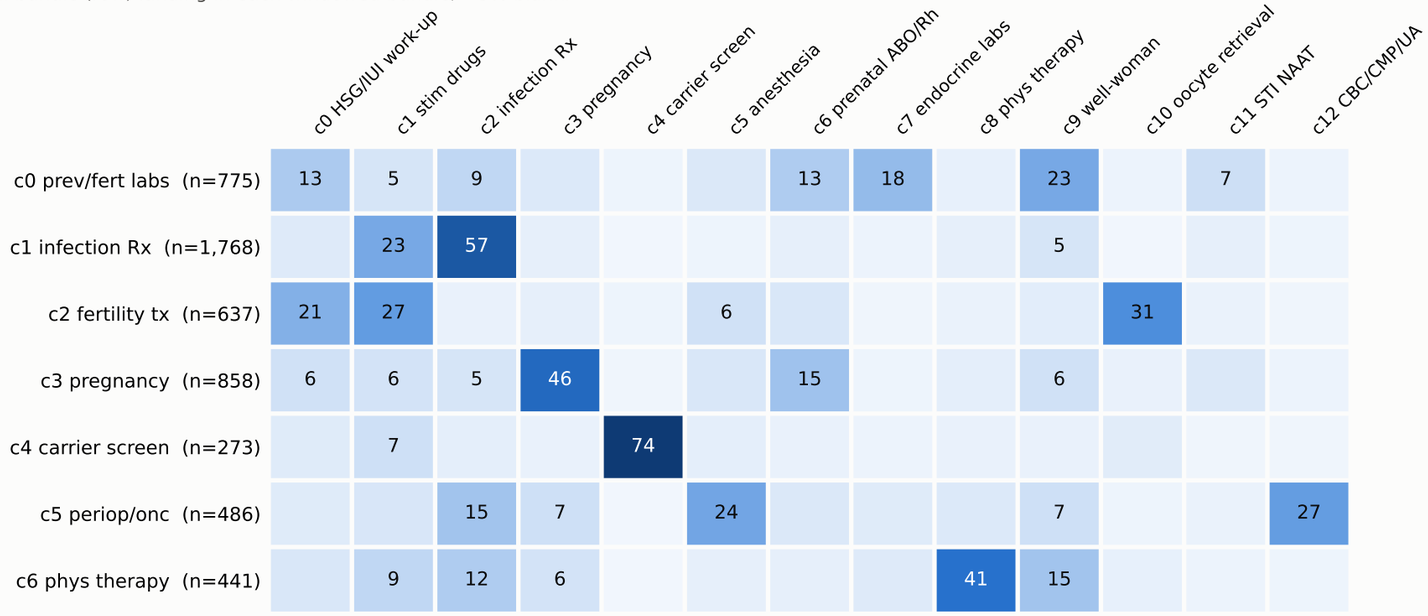
Same 5,238 patients; windowed vocab 2,982 tokens ⊂ full 3,977. Sources: final_table / nmf_components xlsx + .npz patient assignments.

Bundles (with letrozole patterns)	Scored top patterns	Solid verdict	"Same-day bundle" order	Median lift	Largest bundle
full 7 (6)	full 18	full 11 (61%)	full 1	full 3.5×	full c1 infection Rx, 33.8%
win 13 (12)	win 35	win 23 (66%)	win 13	win 5.8×	win c2 infection Rx, 24.1%

Partition agreement at patient level: ARI 0.28, NMI 0.30 (moderate). Only 44% of patients stay with their full bundle's modal windowed bundle.

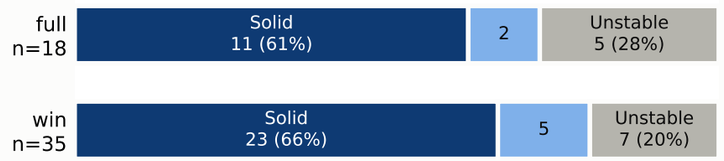
Where does each full bundle go when windowed?

Rows: full k7 bundles. Columns: windowed k13 bundles.
% of each full bundle (row) landing in each windowed bundle; <5% blank.



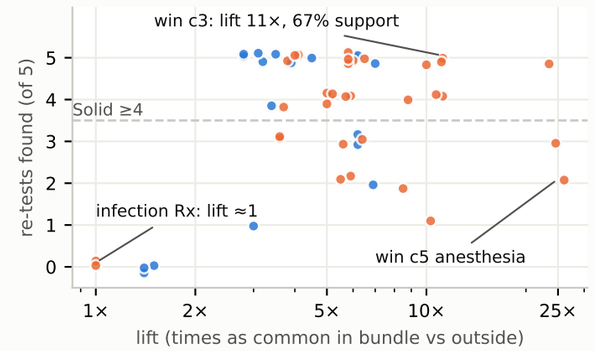
Re-test verdict of top patterns

Top-3 letrozole patterns per bundle. Solid = re-found in ≥4/5 half-splits; middle segment = Suggestive.



Outliers: lift ≠ reliability

All 53 top patterns: full windowed



Similarities

- Same themes recur: fertility, pregnancy, PT, peri-op, labs, infection Rx, carrier screening.
- c4 carrier screening most stable: 92% of win c4 sits in full c4 (273→221 pts, CFTR/HEXA/FMR1 ~98%); 0 letrozole patterns in both runs.
- Infection-Rx bundle is a null in both: only letrozole×N patterns, lift 1.0–1.5, 0/5 re-tests, Unstable.
- Pregnancy (letrozole → US/glucose → live birth) and PT (letrozole → 97110 → 97140) are Solid in both.
- Fertility order unchanged: letrozole → Estradiol → hCG/progesterone, 5/5 re-tests.
- Mostly Solid in both runs: 61% full, 66% windowed.

Differences

- Windowing splits coarse bundles. Full c2 fertility → win c10 oocyte retrieval (83% of c10), c1 stim drugs, c0 HSG/IUI work-up.
- Full c0 labs → win c7 endocrine (68% of c7), c9 well-woman (40%), c11 STI. Full c3 pregnancy → win c3 + c6 prenatal ABO/Rh (40% of c6).
- Lifts rise: PT 6.2→11.2×, pregnancy 7.0→11.2× (±18 mo removes unrelated history from the outside comparison).
- Same-day orderings jump: 13 of 35 patterns vs 1 of 18; order is a tie-break, not evidence.
- Infection Rx shrinks 1,768→1,263; oncology label lost (see outliers).

Outliers & caveats

- win c5 anesthesia: lift 23–26× but outside share ≈1% inflates the ratio; all same-day; 2 of 3 not Solid.
- Full c5 "peri-op/oncology (breast_cancer loads here)": no oncology bundle survives windowing; 78% of win c12 (CBC/CMP/UA) came from it.
- High lift, unstable: full c3 #2 (6.9×, 2/5), win c1 (5.9×, 2/5, 21% support), win c10 hCG (5.5×, 80% support, 2/5), win c12 CMP (8.5×, 2/5).
- Strongest result: win c3 letrozole → US → live birth, 67% of bundle, 11.2×, 5/5.